Empirical Proof Record: VALIDATED

2026-05-20T15:34:51.632175+00:00

Empirical Proof Record — VALIDATED

Proof ID: `edc031e751c9bd300c57c003da8fbc61…`
Created: 2026-05-20T15:34:51.632175+00:00

1. Claim

> Across 100 random (k, n) binomial pairs (seed=20260518, max_n=1000), scipy's "binomtest.proportion_ci(method='wilson')" and statsmodels's "proportion_confint(method='wilson')" produce 95% CI bounds that agree to ≤ 1e-09 on both lower and upper endpoints. (RFC 0002 Phase E1: cross-framework validation at the statistical-primitive layer.)

- **Operationalisation:** Generate N pairs (k, n) with k ~ Uniform(0, n), n ~ Uniform(2, max_n). For each pair, compute the Wilson CI at the configured confidence under both backends. Compute max_abs_diff = max over pairs of max(|low_scipy - low_sm|, |high_scipy - high_sm|). VALIDATED iff max_abs_diff ≤ tolerance.
- **Threshold:** `max_absolute_ci_difference <= 1e-09 proportion`
- **H0:** scipy and statsmodels disagree on the Wilson CI by more than the floating-point tolerance — at least one implementation has a defect.
- **H1:** Both libraries evaluate the same closed-form Wilson CI and produce identical bounds to floating-point tolerance.

2. Verdict

VALIDATED — observed `3.3306690738754696e-16`
100 pairs checked; max |low_scipy - low_sm| = 1.110e-16; max |high_scipy - high_sm| = 3.331e-16; max overall = 3.331e-16 (≤ 1e-09 tol)

3. Evidence

Pillar	Statistic	Value	95% C
wilson_ci_scipy_vs_statsmodels	max_absolute_ci_difference	3.3306690738754696e-16	-

4. Pre-registration

- Registered at: `2026-05-20T15:34:51.377070+00:00`
- Config hash: `ab8fb015887c4a2ac85c8b2ffe1e676a…`
- Data hash: `ab8fb015887c4a2ac85c8b2ffe1e676a…`

5. Signature

`bd67a5cbe9137995f29b6ad7959d77b56da16dbeaf811a5a57bcffe40a24443e`